STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: CNC-01 | TAG: 01168375

KIA 3-Axis CNC Vertical Machining Center

Facility Zone: Zone 2: CNC Machining (CNC) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

CRITICAL ENTANGLEMENT WARNING: STRICTLY NO GLOVES WHILE SPINDLE / BLADE IS ROTATING

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses
2. Closed-Toe Shoes
3. NO GLOVES When Loading Spindle or Near Rotating Parts

2. PRIMARY OPERATIONAL HAZARDS

1. Side-Mount Tool Changer (SMTC) loses 3" Z-clearance during tool swing
2. High-speed flying tool/workpiece ejection hazard (8,000 RPM inline direct-drive spindle, 600 IPM rapids)
3. Automatic axis motion during program execution

3. BEFORE OPERATION CHECKLIST

1. Machine Specs: 8,000 RPM inline direct-drive spindle, 600 IPM rapids.
2. Do NOT operate above 6,000 RPM without balanced tooling.
3. Verify tool length offsets (Z-heights) and workholding clearance (account for 3" Z-drop during SMTC swing).

4. DURING OPERATION SAFETY RULES

1. Keep one hand on E-Stop during first run prove-out at 5% rapid override.
2. Keep enclosure doors closed during all machining operations.
3. Monitor Distance-To-Go readout before tool approach.

5. AFTER USE & SHOWROOM CLEANUP

1. Brush chips ONLY when machine is powered down.
2. NEVER use compressed air inside enclosure (forces particulates under way covers and into spindle bearings).
3. Wipe down windows and return toolholders to rack.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Slam red E-Stop mushroom button on front CNC control pendant immediately; do not open doors until spindle comes to a full stop.